

Hybrid Agile-Waterfall Methodology for Philippine Public Sector Procurement Planning System

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Abstract

Procurement modernization in the Philippines has been instituted in 2003 by the Government Procurement Reform Act or RA 9184 and was further enhanced by the 2024 New Government Reform Act or RA 12009. However, despite reforms, procurement systems still need to hurdle transparency issues and regulatory gaps causing delays in the public sector supply chain and requirements. Digital tools and innovative processes need to be integrated into the systems to provide equitable budgetary allocations and increase the level of public confidence in government operations. Equally important is the significant considerations on regulatory compliance and operational efficiency when developing public procurement systems. The procurement planning stage is a critical phase in the procurement system because end users use this to determine their needs and requirements in the performance of their mandated roles and functions. This research presents contextual adaptation of a software development framework uniting Agile and Waterfall methodologies in creating the public procurement planning system. Specifically, in the procurement planning context, this approach supports regulatory requirements and adaptability with end user needs gathered through various stakeholder engagements throughout the development cycle. The design process spanned eight weeks by combining regulatory reviews with interviews and focus groups to create the requirements followed by rounds of prototyping, coding, and testing activities. The system design includes major features to create procurement plans (PPMP/APP), to assist end users in their market scoping activity documentation, and to gather insights for decision-making into procurement activities of the agency via customized dashboard serving as an analytical tool. Pilot tested in a state university, the system proves its ability to comply with the Philippine procurement rules specifically in the creation of procurement plans. Testing results show high acceptability with regard to scalability, consistency and reliability, and bandwidth and size using Apache JMeter. Google Lighthouse assessment of the quality of webpages resulted in 96% for performance, 82% for accessibility, 100% for best practices, and 91% for SEO. A UTAUT questionnaire with a generally acceptable reliability and validity (Cronbach's Alpha ≥ 0.70) was used for evaluation and resulted in a grand mean of 4.56 reflecting the system's acceptability and utility. The hybrid approach provides an adaptable model to improve procurement planning activities in the Philippine public

sectors. The integration of applicable emerging technologies into similar systems using the hybrid Waterfall-Agile software engineering methods can be explored in future research to achieve better public sector outcomes.

CCS Concepts

• IT governance; • Web-based interaction; • Real-time systems software;

Keywords

E-governance, hybrid Waterfall-Agile, procurement planning, software engineering, system design

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1 INTRODUCTION

The procurement sector is undergoing a significant transformation, necessitating the alignment of strategies with organizational strategies, sustainability, and operational efficiency objectives, where value for money and compliance are integral to strategic planning, needs identification, supplier selection, contract management, and performance evaluation. Under the significant procurement planning phase, a meticulous assessment of user requirements and market conditions, along with risk evaluation, is needed for informed decision-making [1], [2]. The success of the procurement planning – the initial phase, determines the success of succeeding phases (e.g. identification of modes of procurement, conduct of procurement, monitoring, and supplier performance assessment).

The procurement system in the Philippines shows signs of fragmentation, operational inefficiency, and reduced transparency due to non-compliance problems despite ongoing reform efforts [3]. Public entities need to use manual processes with forms and spreadsheet data entry because their automated systems do not have the regulatory requisites, thus the need for system development, and governance to improve procurement results.

The Government Procurement Reform Act (RA 9184) serves as the primary legislation governing procurement processes in the Philippines. The procurement process includes procurement planning, bidding, and awarding, as well as contract management and monitoring [4], [5], [6]. In 2024, the New Government Procurement Act (RA 12009) was enacted, amending RA 9184 to augment public procurement's transparency, accountability, competitiveness, and value for money [7].



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This study introduces hybrid Agile-Waterfall model to assist public sectors in developing regulatory-compliant procurement planning system, a significant phase in the public procurement processes – while utilizing the Agile’s adaptability to optimize procurement processes [8]. This initiative presents a hybrid system development strategy in the context of a procurement planning system for government agency adoption and generates data-driven insights and workflow improvements for stakeholders.

1.1 Background of the Study

The implementation of procurement systems leads to better transparency, accountability, equal treatment, and sustainable practices. Advance technologies integrated into procurement systems enhance procurement efficiency by mitigating corruption and enhancing decision-making capabilities [9]. The OECD benchmarks and the UN Commission on International Trade Law (UNCITRAL) [10] serve as models to modernize and standardize procurement through digital systems.

The Philippine laws on procurement requires public organizations to submit procurement plans with market scoping documentations using approved standard forms at the procurement planning phase. Procedures not complying with the standard process (as required) lead to inefficient procurement planning outcomes, further leading to procurement delays, and delays in service provision.

This study offers a hybrid development approach for procurement planning systems complying with the needs and requirements of the law and the stakeholders. Requirements such as the use of structured templates for PPMP/APP creation and submission, a market scoping assistant to help end users in the market scoping documentation (price and supplier), and dashboard module to provide insights for decision-making are integrated into the system. This public procurement planning system enables public agencies to automate their procurement operations, which results in faster operations and lower expenses.

Other countries in the region have advanced digital procurement systems such as Malaysia’s MyProcurement portal which empowers SMEs through digital access [11], and Singapore’s Doxa Connex platforms which streamlines procurement [12]. These initiatives emphasize the use of technologies which Philippines can follow.

1.2 Problem Statement and Research Objectives

The Philippine government agencies are using fragmented, and uncoordinated public procurement planning system leading to project delays. Non-compliance issues to the New Government Procurement Act (RA 12009) results in poor management of public funds. This gap requires an advanced solution providing user-friendly procurement planning system that maintains compliance standards delivering results with full transparency and accountability.

This study addresses the identified problems by exploring the following research questions:

What are the key functional, non-functional, and compliance requirements for an effective procurement planning system in the Philippine public sector?

How can a hybrid waterfall methodology be effectively employed to design and develop a procurement planning system that meets these requirements?

2 LITERATURE REVIEW

Software development uses Agile and Waterfall as two separate methodologies which bring different advantages and constraints to the development process. Waterfall represents a linear approach which works best for projects that need to follow set requirements while emphasizing [13], [14]. This methodology serves to provide benefits in stable conditions where changes are restricted, and adherence to compliance is important [15]. The Agile approach offers flexible frameworks that use iterative methods to support stakeholder participation and project adjustments for changing requirements, making it suitable for projects with rapid changes [13], [16]. The hybrid strategy seeks to bring together the benefits of two methodologies, uniting Waterfall’s organized planning and the versatile nature of Agile [13], [16] particularly beneficial in complex projects, offering improved project outcomes, accelerating delivery, and enhancing user satisfaction [13], [17]. Research indicates hybrid models appear more frequently in public systems and projects which require a combination of waterfall predictability and Agile flexibility [15].

2.1 Development in Hybrid Methodologies

The implementation of hybrids requires developers to perform repeated coding tasks throughout Waterfall gates which operate within Agile sprint cycles. Research conducted in public sector environments demonstrate that hybrid systems fulfill regulatory requirements by enabling users to provide feedback [18].

2.2 Application of Data-driven Technologies in Public Procurement Processes

Research has shown that data-driven technologies used in public procurement bring various advantages and difficulties to the process. Machine learning models achieve high precision in tender variation prediction which helps public contracts finish faster and cheaper [19]. The role of AI in procurements is increasingly recognized as important for enhancing efficiency and data security and its integration into administrative activities [20]. Research activities have increased because technology-based public procurement systems play a vital role in supporting effective governance and economic development [21]. AI serves as a solution to combat corruption while improving procurement efficiency in Nigerian government [22].

2.3 Integration of Analytics into Public Procurement Frameworks

Previous studies have examined how analytics integration into public procurement systems would boost operational performance and strengthen oversight mechanisms. One study demonstrates how data-based frameworks help organizations achieve better procurement results [23]. Organizations can achieve better transparency through advanced data mining techniques in aid to developing sourcing plans for consolidating similar goods and services across different procurement systems [24]. The role of big data analytics revealing trends as exclusive commercial relationships may indicate procurement fraud [25]. Business analytics combined with analytics

and performance management through a framework demonstrated its success in public sector organizations [26].

2.4 Challenges in Adopting e-procurement Systems

Multiple barriers exist during e-procurement system implementation because they affect technological elements, organizational systems, and regulatory requirements. The integration of information systems with security measures and IT infrastructure availability stands a major technological challenge [27]. Employee resistance to change, lack of suitable skills, and inadequate leadership backing [28], [27] are organizational challenges. The adoption of new technologies is also faced with legal compliances, lack of proper guiding frameworks and policies that hinder implementation [27]. Despite challenges, many organizations recognize the need for increased usage and the development of strategies to overcome said barriers [29].

3 METHODOLOGY

The hybrid Agile-Waterfall methodology organizes project phases through Waterfall's sequential plan-driven approach and Agile's flexible iterative structure. The hybrid model combines Waterfall's project scope definition with Agile's adaptability to create a method that supports detailed planning of initial phases [13], [16]. The project lifecycle benefits from Agile adaptability because teams can make changes and receive stakeholder feedback at different stages. The hybrid approach uses Scrum and Agile method for development work to achieve fast iteration cycles and progressive delivery while keeping Waterfall's organized structure for other project stages [30].

The methodology suits Philippine public procurement planning system because it combines Waterfall's regulatory compliance features with Agile's ability to adapt to changing project conditions.

Requirements were gathered through stakeholder interviews and focus groups involving end-users, procurement officers, and administrators of a government agency. This study was done and piloted in a state university environment where procurement activities are governed by the existing procurement laws. Requirements were identified, prioritized, and validated linking to design and tests.

4 RESULTS AND DISCUSSION

The requirements gathering activities engaged stakeholders through interviews and focus groups, review of the regulatory requirements provided by RA 9184 and RA 12009, and technical consultations with experts helped with the process identification. Figure 1 shows the use case flowchart of the procurement planning system. As shown, the requesting officer, procurement officer, and administration are primary actors distinguishing the three primary actors. It also exhibits the system's core functionality supporting automated and data-driven procurement planning.

The creation of the procurement plan by the requesting officer is at the heart of the flowchart enabling end users to perform procurement item searching, adding and editing, with access to the market scoping module – a mandatory requirement of RA 12009.

The completed procurement plan is then saved and submitted for evaluation by the procurement officer.

The system is used to evaluate procurement plans by an assigned committee where modifications can be made prior to approval. Once approved, PPMPs are then consolidated into the agency's annual procurement plan (APP). Items are provided with pricing forecasts using data gathered from 2017.

To supplement procurement planning scenario is the procurement planning process flow shown in figure 2 showing transparent stakeholder interactions to deliver results.

The call for requirements (creation of PPMP) is to be initiated by the administration following budget and procurement cycles of the agency in sync with the national budget calendar. This means that the agency needs to schedule properly the PPMP/APP creation to comply with national level submissions.

A RACI matrix of the procurement has been created from the tasks identification during requirements gathering shown in table 1. It defines the roles and responsibilities of the requesting officer, procurement officer, and the administration as to who performs the task, who approves and answerable for the task, who provides inputs or expertise, and who is informed of progress or outcomes.

4.1 Functional and Non-functional Requirements

Functional requirements (FRs) outline specific capabilities and behaviors of the procurement planning system. These are identified to support the creation, and management (including evaluation) of procurement plans in compliance with the Philippine procurement laws. Table 2 below shows the functional requirements to ensure alignment with user needs and regulatory standards. Requirement IDs are created corresponding to the requirements.

The FRs are validated during gathering of requirements and tested as part of the development phase outlined in the methodology.

Table 3 identifies the non-functional requirements (NFRs) to ensure it meets operational, regulatory, and user expectations apart from the core functionalities. These NFRs are based on software engineering principles and tailored to the system's contexts as a web-based compliant procurement planning tool.

The NFRs are validated during gathering of requirements and are evaluated during testing and UTAUT evaluation. NFRs define how the system performs under real-world constraints (beyond mere functionality) and mitigate the risks in operational failure, non-compliance, and inefficiency which are critical in public sector environments governed by regulations such as the RA 9184 and RA 12009.

4.2 Hybrid Agile-Waterfall software development framework

Figure 3 shows the project management methodology used in designing the procurement planning system combining the structured, sequential nature of Waterfall with the iterative, flexible characteristics of Agile. The methodology commences with comprehensive research, succeeded by the initiation phase and the development of the conceptual framework. Then detailed requirements planning and analysis, user-centric design, prototyping, and eventual

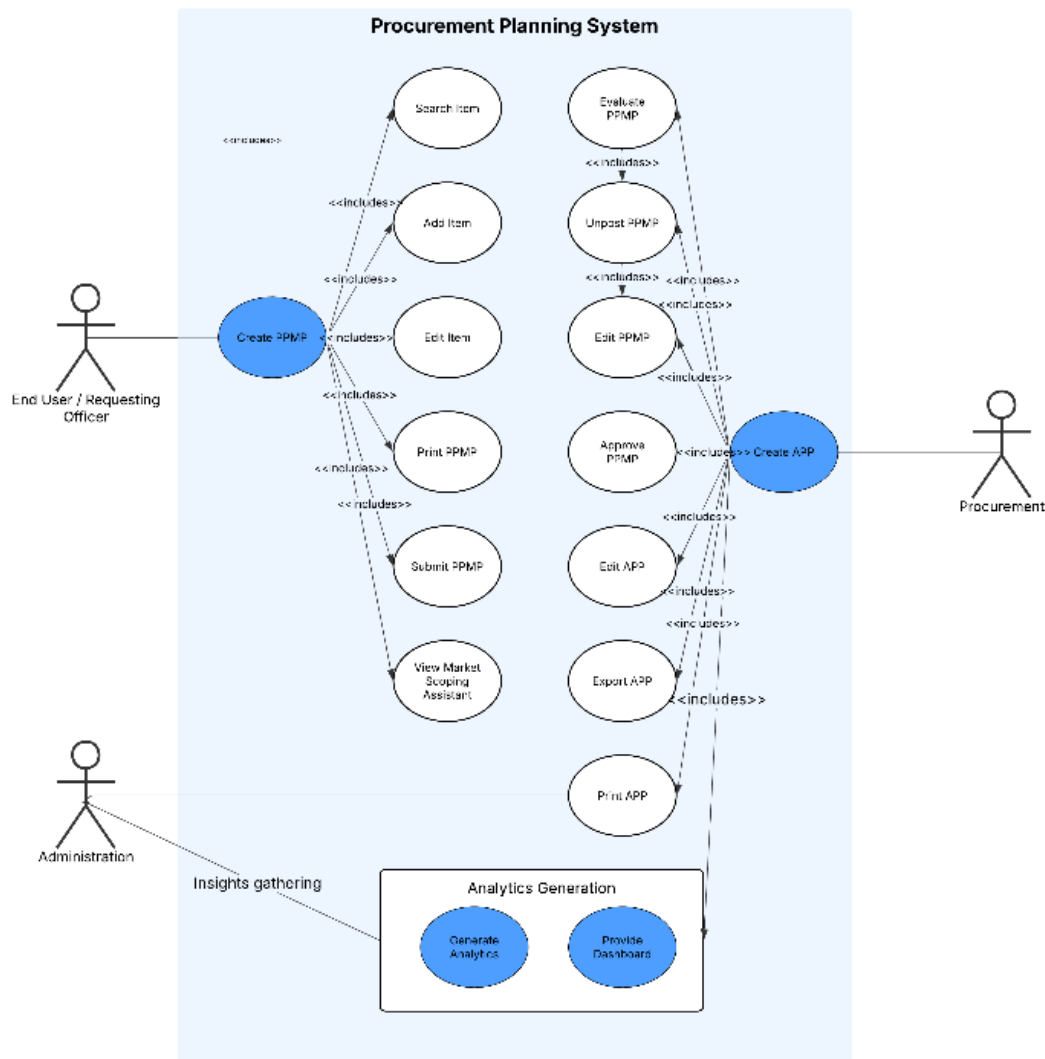


Figure 1: Procurement Planning System Use Case Diagram

Table 1: Procurement Planning RACI Matrix

Task	RO	PO	A	SA	Task	RO	PO	A	SA
Create/Edit PPMPs	R	C	I	I	Send Automated Notifications	I	R	I	A
Utilize Market Scoping Assistant	R	C	I	I	Configure Notification Preferences	I	I	I	R
Submit PPMPs	R	A	I	I	User Registration, Authentication, Role Assignment	I	I	I	R
Consolidate PPMPs into APP	I	R	A	I	Account Management	I	I	I	R
Generate APP Reports	I	R	A	I	Generate Analytics	I	R	A	I
Import/Export Data	I	R	I	C	Provide Visual Dashboards	I	R	A	I

Legends: RO-Requesting Officer, PO-Procurement Officer, A-Administration, SA-System Administrator, R-Responsible, A-Accountable, C-Consulted, I-Informed.

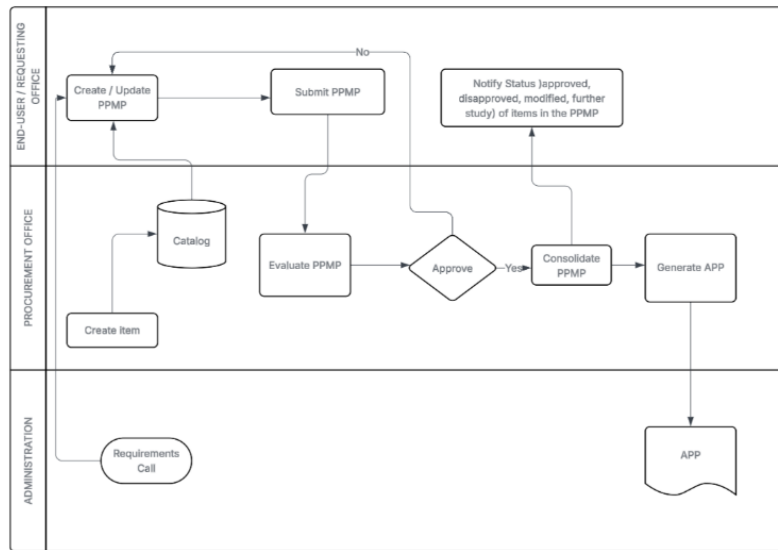


Figure 2: Procurement Planning Process Flow

Table 2: Functional Requirements of the Procurement Planning System

Category	Requirement ID	Description
User Management	FR-UM-01	Allow user registration, authentication, and role assignment (e.g., procurement officer, auditor) for access control.
PPMP Creation	FR-UM-02	Enable password reset and account management for authorized users.
	FR-PC-01	Allow users to create and edit PPMPs with fields for project details, procurement items, and timelines.
Analytics	FR-PC-02	Support the market scoping requirement of the RA 12009 via the market scoping assistant
Generation	FR-AG-01	Generate analytics (e.g., price forecasting, supplier performance) based on historical data.
	FR-AG-02	Provide visual dashboards for analytics insights (e.g., trends, status).
APP	FR-AC-01	Consolidate multiple PPMPs into a single APP with automated data aggregation.
Consolidation	FR-AC-02	Generate APP reports formats compliant with standard forms.
Data	FR-DM-01	Import/export procurement data from/to external systems
Management	FR-DM-02	Maintain an audit trail of all procurement planning processes (e.g., edits, submissions).
Notification	FR-NT-01	Send automated notifications (e.g., submission deadlines) to users.
	FR-NT-02	Allow users to configure notification preferences.

construction follow. Each phase is delineated by checkpoint to guarantee systematic advancement. After construction, the flow transitions into a cyclical loop involving integration (and testing), deploy, and support/maintenance, which then loops back to construction to allow continuous refinement and responsiveness to feedback, embodying Agile principles within a Waterfall framework.

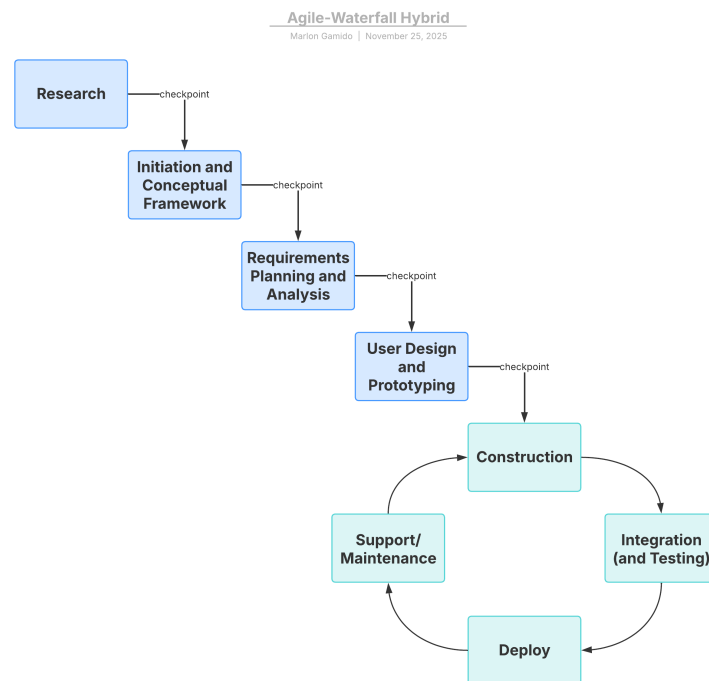
This flow mirrors the procurement planning system development phases. The research phase corresponds to the literature review, stakeholder interviews, process observation and analysis of public procurement regulatory requirements – RA9184/RA12009. The initiation and conceptual framework phase involves defining the project scope, legal compliance, and drafting the conceptual framework. These are gathered from developmental perspectives

of stakeholders and public procurement directions in the Philippines. Requirements, planning, and analysis includes gathering functional and non-functional requirements, creating the software requirements specification (SRS), and mapping out current versus proposed processes. Software engineering practices are adopted to help in the identification of functional and non-functional requirements of the system. The user design prototyping phase focuses on wireframes, personas, and iterative prototypes development using applicable tools. Construction involves backend and frontend development, analytics engine integration, and compliance feature implementation.

The cyclical loop of integration, development, and support/maintenance is reflected in the system through continuous testing, pilot deployment, user acceptance testing (UAT), and iterative refinements based on feedback. These guarantee that the system is flexible

Table 3: Non-Functional Requirements of the Procurement Planning System

Category	Requirement ID	Description
Performance	PERF-01	Process user requests (e.g., plan generation) with minimal latency to support efficient procurement planning workflows.
Reliability	REL-01	Maintain high availability for critical procurement planning operations, including PPMP and APP consolidation.
Usability	USAB-01	The user interface shall be intuitive for procurement officers, budget analysts, and administrators with varying technical expertise.
Security	SEC-01	Protect sensitive procurement data (e.g., supplier details, budgets) from unauthorized access and breaches.
Maintainability	MAINT-01	The system architecture shall facilitate updates, bug fixes, and enhancements without disrupting service.
Scalability	SCAL-01	Accommodate growth in university procurement volume (e.g., increasing projects or users).
Interoperability	INTER-01	Integrate with existing university systems (e.g., budget system, external APIs for user and supplier data).
Compliance & Auditability	COMP-01	Ensure all processes align with regulatory requirements for transparency and accountability.

**Figure 3: Hybrid Agile-Waterfall software development framework for the procurement planning system**

and reactive to user needs while considering regulatory requirements.

Table 4 shows the development design of the hybrid methodology spanning 8 weeks and focusing on structured but iterative process of the procurement planning system. It details the system’s 8-week development under a hybrid methodology, demonstrated by

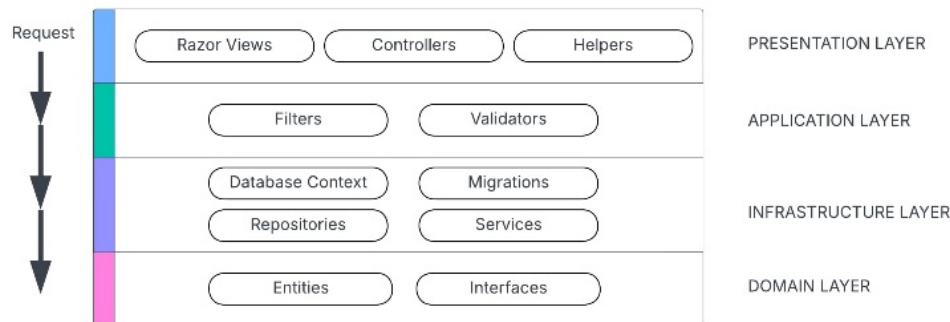
the system-user interactions with market scoping assistant and analytics engine’s development.

The development schedule is as follows:

Week 1 involves project initiation with scope definition, legal compliances, and an initial framework gathered through a stakeholder kickoff. Activities include reviewing RA 9184/ 12009, best practices, and drafting the conceptual framework.

Table 4: Design of the System Development Schedule Following the Hybrid Methodology (Objectives)

Week	Phase	Objective
1	Initiation and Conceptual Framework	Define scope, legal compliance, and initial framework
2	Requirements Planning and Analysis	Gather and analyze requirements
3-4	User Design and Prototyping	Design and iterate prototypes
5-6	Construction	Build and integrate analytics
7	Deployment and Testing	Deploy and test the system
8	Evaluation and Handover	Evaluate performance and hand over

**Figure 4: Procurement Planning System Architecture**

Week 2 involves gathering and requirements analysis from interview and focus groups with stakeholders, analyzing historical data from 2017 to 2025, and mapping current and proposed procedures.

Weeks 3-4 focus on user design and prototyping, and creation of mockups using tools like Replit with continuous user feedback gathering. A minimalist user interface design is adopted to provide similar character with other organizational system interfaces.

In weeks 5-6, construction was done with frontend and backend construction (including API integrations from HR system and google for market scoping requirement), analytics engine implementation using ML.net, and integrating the compliance features from PPMP to APP creation.

Week 7 covers deployment and testing which includes pilot deployment, unit, integration, and User Acceptance Testing (UAT) with refinements based on feedback.

Week 8 covers evaluation and handover where implementation significantly contributes to the usability of the procurement planning system.

4.3 System Architecture

The system architecture shown in figure 4 is optimized for the procurement planning system compliant with RA 9184 and RA 12009. The presentation, application, infrastructure, and domain layers promote separation of concerns, dependencies, and extensibility with the hybrid Agile-Waterfall approach.

Table 5 shows the procurement planning system architecture layers with their corresponding descriptions. It includes the layer's role with corresponding identified components to perform their tasks.

4.4 Network Architecture

The network diagram shown in figure 5 shows how various components of the system are interconnected across the public agency's infrastructure. It shows the flow of data and communication between users, system modules, databases, and external services to ensure a secure and efficient procurement planning environment.

4.5 Analytics Implementation

The analytics engine uses the ML.NET on procurement data collected in the university from 2017 to 2025. ML.NET is a machine learning framework developed by Microsoft that allows .NET developers to build, train, and deploy analytics modules. This provides machine learning abilities into this .NET application without needing external services. Figure 6 below shows a sample code using the ML.NET.

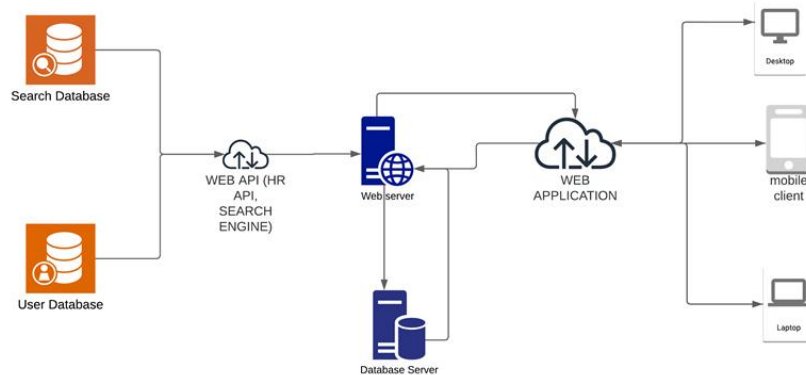
4.6 System Features

The market scoping assistant is included, which provides real-time supplier specifications and price estimates (whenever available and posted) via Google API queries, helping users comply with RA 12009 documentation requirements.

The dashboard design includes interactive charts for insight gathering for decision support. The user can explore annual budgetary distribution of procurements categorized as office equipment, furniture and fixtures, office supplies, accountable forms, medical/dental supplies, gasoline, oil, and lubricants, textbooks and reading materials, construction materials, cleaning materials, sports and cultural supplies, semi-expendables, and IT accessories. The system can add categories as needed. It can also display the allotted budget per

Table 5: Procurement Planning System Layer Composition

Layer	Description
Presentation	Interacts directly with users and forwards processed requests to lower layers. It is composed of razor views responsible for rendering templates with dynamic data using ASP.NET Core, controllers acting as entry points for http requests, and helpers which assists in view rendering. ASP.NET core is used in this development to be consistent with the locale’s framework for building cross-platform web applications.
Application	Acts as the mediator, enforcing regulations and facilitating coordination between the presentation layer and the subordinate layers. It governs logic that is specific to the application. The layer contains authentication and authorization filters which performs cross-cutting operations and logging functions and input validation that checks data integrity before processing.
Infrastructure	Demonstrates practical uses for data storage and retrieval operations which handle external dependencies and data persistence and technical implementation complexities. It consist database context that oversees database connections, migrations that guarantee versioned alterations to the database architecture, repositories that furnish abstraction frameworks for data access and services.
Domain	Encapsulates the system’s business logic. It is the core of the system which contains business entities and rules independent of technology. It is composed of entities with properties and behavior, and interfaces which define roles for services.

**Figure 5: Procurement Planning Network Architecture**

```

1 reference
public class CostPredictController : Controller
{
    private readonly RegressionService _regression;
    0 references
    public CostPredictController(RegressionService regression)
    {
        _regression = regression;
    }
    0 references
    public IActionResult Index()
    {
        return View();
    }
    [HttpGet]
    0 references
    public IActionResult Get(float quantity, float rating, string material)
    {
        var cost = _regression.Predict(quantity, rating, material);
        return Ok(new { Quantity = quantity, Rating = rating, Material = material, PredictedCost = cost });
    }
}

```

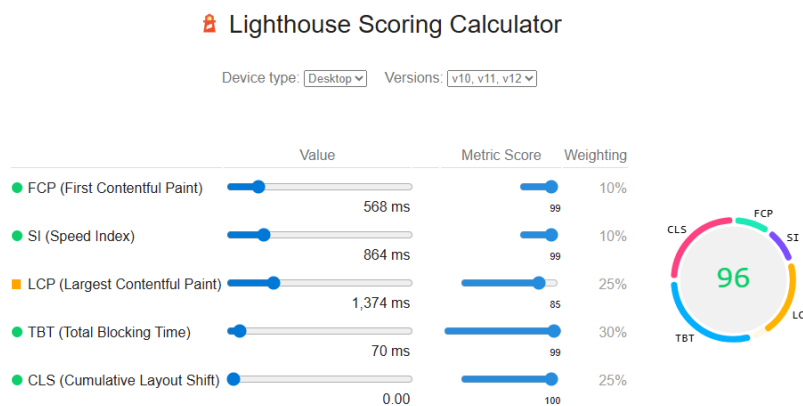
Figure 6: ML.net sample system code

office and the budget used for procurements to identify the budget utilization rate of each office, a measure of end-users’ spending versus budget allotment.

The Annual Procurement Plan (APP) of the agency is one of the significant outputs of the procurement planning system. It is a requirement of the Philippine government to submit the agency’s

Table 6: Load and Spike Test Results Using Apache JMeter

Endpoint	Avg Load (ms)	Std. Dev. Load	Avg. Spike (ms)	Std. Dev. Spike	Avg increase
GET Login Page	3	2.68	7	1.49	133.3
POST Login	145	7.53	156	12.02	7.6
HomePage	5	0.72	4	0.74	-20.0
PPMP	5	0.83	4	0.84	-20.0
Dashboard	3	1.10	3	0.74	0.0
PPMP Evaluation	4	0.83	4	0.90	0.0
CSE Evaluation	4	0.74	4	0.91	0.0
APP	4	0.93	4	0.89	0.0
Budget Requirement	4	0.66	4	0.92	0.0
Employee	3	0.73	3	1.19	0.0
Office	3	0.64	3	0.74	0.0
APPSetting	3	0.61	3	0.76	0.0

**Figure 7: Web performance audit using Google Lighthouse**

APP for budgeting purposes and for other decision-making activities. A printed output following the standard forms as required can be produced.

5 EVALUATION

The procurement planning system was subjected to two-scenario tests – the load test simulating steady user traffic with 100 samples per endpoint, and a spike test simulating a sudden traffic surge with 200 samples per endpoint using the Apache JMeter – an open-source software test tool. Twelve (12) endpoints were the subjects of the tests where key metrics measured include response times (ms), standard deviation for response consistency, error rate, throughput (requests per second), bandwidth usage (received/sent KB/sec), and average response size (bytes).

Table 6 shows the summary of results of the load and spike tests using Apache JMeter. Results show that the system is resilient considering scalability, consistency and reliability, bandwidth and size. Scalability was evident when the spike test doubles the samples but response times increase minimally implying effective resource

utilization. Consistency and reliability is demonstrated by low standard deviation in both tests indicating predictable performance. Bandwidth and size shown by larger endpoints (e.g., HomePage at approximately 38K bytes) scale bandwidth usage linearly without issues. However, sent KB/sec increases more in spike for POST Login due to heavier requests payload. Bottlenecks exist in POST Login consistently underperform with the largest increase in response time and variability.

A web performance audit was conducted using Google Lighthouse. This tool is used to audit performance, accessibility, best practices, and SEO. Figure 7 below shows the system's performance rating of 96%.

Aside from performance, the system got 82% for accessibility, 100% for best practices, and 91% for SEO.

The procurement planning system received evaluation from thirty (30) respondents including requesting officers, procurement officers, and administration personnel through the Unified Theory of Acceptance of Use of Technology (UTAUT) questionnaire. The UTAUT questionnaire was also subjected to reliability and validity assessment using Cronbach's Alpha targeting a computed value

Table 7: UTAUT Reliability and Evaluation Result

UTAUT Criteria	Cronbach's Alpha	Mean Score
Performance Expectancy (PE)	0.73	4.70
Effort Expectancy (EE)	0.81	4.60
Facilitating Conditions (FC)	0.90	4.40
Social Influence (SI)	0.87	4.50
Behavioral Intention (BI)	0.72	4.60
Grand Mean		4.56

of ≥ 0.70 as scores are shown in table 7. Reliability scores above 0.70 generally indicate an acceptable reliability for each UTAUT construct in assessing the internal consistency of questionnaire items within each category.

Also shown in table 7 is the deep agreement of the participants about the system acceptability and their plans to use the system. Evidently, the participants strongly accepted the system because their average rating of 4.56 indicates they found it both useful and easy to use. The system's performance expectancy (PE) scores demonstrate its ability to operate efficiently while supporting informed and analytical decisions. The system requires minimal effort expectancy, but training can be extended to overcome their initial learning difficulties as evidenced in the facilitating conditions (FC) criterion. The results also show that social influence (SI) plays a key role because stakeholders actively support the system, leading to higher adoption rates. The behavior intention (BI) demonstrates that users will continue to use the system because of high confidence in its operation which support long-term sustainability in Philippine public sector operations.

6 LIMITATIONS OF THE STUDY

The scope of this study is limited to the development of a web app of a procurement planning phase of the public procurement system in the Philippines. This provides creation of the annual procurement plan of the organization from the evaluated project management plans procurement plans of respective end-users. It also provides a market scoping documentation assistant with integrated web search API and a dashboard to visualize procurement plans to get real-time insights for decision-making. The output of this system will be used for actual procurement activities.

7 CONCLUSION

The research successfully utilized a hybrid Agile-Waterfall methodology addressing the need for regulatory compliance (creation of agency's PPMP/APP with market scoping) and the adaptive development in procurement planning system. Further concludes:

The system's design successfully incorporates functional, non-functional, and regulatory requirements fit for the Philippine public sector procurement planning processes with active engagement of stakeholders considered as key to requirements gathering and development.

The hybrid method provides better predictability for regulations, and the 8-week development is feasible for similar systems and as evidenced by the results of various tools including Apache JMeter, Google Loghouse, and the UTAUT evaluation. This offers

a system development framework that can be adopted in similar contexts such as procurement planning systems.

8 KEY CONTRIBUTIONS

This study introduces a hybrid framework by merging Agile and Waterfall strategies in software development specifically applied in the development of public procurement planning systems which balances legal requirements with iterative development. The success of this development is evidence enough to adopt this methodology in the public procurement planning management context.

It integrates a dashboard anchored on data to facilitate informed decision-making. Further, it offers a reproducible design blueprint for other government entities to implement similar systems.

References

- [1] M. Cooper, "Navigating Complexity: Insights into Strategic Procurement Practices in Global Supply Chains," Jul. 08, 2024. doi: 10.20944/preprints202407.0590.v1.
- [2] J. Kodym, "Buying what matters: towards a value model allowing to implement policy preference in public procurement," *Journal of Public Procurement*, vol. 24, no. 4, pp. 465–477, Oct. 2024, doi: 10.1108/JOPP-05-2024-0052.
- [3] D. Jones, "Procurement reform in the Philippines: The impact of elite capture and informal bureaucracy," *International Journal of Public Sector Management*, vol. 26, May 2013, doi: 10.1108/IJPSM-05-2013-0068.
- [4] J. A. Tanghal, and A. M. Navarro, "The promises and pains in procurement reforms in the Philippines," *RePEc: Research Papers in Economics*, Sep. 2017, [Online]. Available: <https://www.econstor.eu/bitstream/10419/173593/1/pidsdps1716.pdf>
- [5] K. D. E. Gabiana, M. A. N. Polinar, and G. C. H. Baquero, "The Bidding Process of The Cebu Provincial Government: A Case Study," *International Journal of Multidisciplinary: Applied Business and Education Research*, vol. 4, no. 5, pp. 1456–1466, May 2023, doi: 10.11594/ijmaber.04.05.07.
- [6] A. B. Yetyet, and A. S. Escalona, "Problems affecting procurement performance in Metro Kalibo Water District," *Deleted Journal*, vol. 5, no. 1, pp. 206–212, Sep. 2024, doi: 10.53378/iarr.924.129.
- [7] GPPB, "GPPB Resolution No. 02-2025 (Approving the Implementing Rules and Regulations of Republic Act No. 12009, Otherwise known as the New Government Procurement Act)," Accessed: Nov. 24, 2025. [Online]. Available: <https://www.gppb.gov.ph/wp-content/uploads/2025/08/GPPB-Resolution-No.-02-2025.pdf>
- [8] I. E. Castañeda, Ma. C. C. Cuntapay, L. R. B. De Guzman, A. M. G. de Vera, A. R. L. Toledo, and C. L. De Leon, "The Government Procurement Law (RA 9184) And Its IRR: Flaws on the Compliance Based on the Observations in the Procurement Bidding Process by the Employees of the Philippine Center for Postharvest and Mechanization Development (PHilMech)," *International Journal of Advanced Engineering, Management and Science*, vol. 9, no. 5, pp. 07–14, 2023, doi: 10.22161/ijaems.95.2.
- [9] M. Mircea, M. Stoica, and B. Ghilic-Micu, "Analysis of the Impact of Blockchain and Internet of Things (IoT) on Public Procurement," *IEEE Access*, vol. 10, pp. 63353–63374, 2022, doi: 10.1109/ACCESS.2022.3182656.
- [10] S. L. Schooner and C. R. Yukins, "Public Procurement: Focus on people, Value for Money and Systemic Integrity, Not Protectionism." [Online]. Available: [http://ssrn.com/abstract=\\$1356170https://ssrn.com/abstract=\\$1356170Electroniccopyavailableat:https://ssrn.com/abstract=\\$1356170](http://ssrn.com/abstract=$1356170https://ssrn.com/abstract=$1356170Electroniccopyavailableat:https://ssrn.com/abstract=$1356170)
- [11] K. K. Soong, E. M. Ahmed, and K. S. Tan, "Factors Affecting Malaysia's SMEs in Using Public Electronic Procurement," *Journal of Information and Knowledge Management*, vol. 19, no. 2, Jun. 2020, doi: 10.1142/S0219649220500082.

- [12] Doxa Holdings International Pte. Ltd, "The All-in-One Platform that Manages your Procurement, Payments & Financing." Accessed: Oct. 27, 2025. [Online]. Available: <https://www.doxa-holdings.com/>
- [13] O. G. Eyieyien, C. Idemudia, P. O. Paul, and T. I. Ijomah, "Advancements in project management methodologies: Integrating agile and waterfall approaches for optimal outcomes," *Engineering Science & Technology Journal*, vol. 5, no. 7, pp. 2216–2231, Jul. 2024, doi: 10.51594/estj.v5i7.1312.
- [14] S. Gupta *et al.*, "Evaluating Waterfall vs. Agile Models in Software Development for Efficiency and Adaptability," 2024, pp. 142–148. doi: 10.4018/979-8-3693-3318-1.ch008.
- [15] A. Rahman, "Agile Project Management: Analyzing the Effectiveness of Agile Methodologies in its Projects Compared to Traditional Approaches," *Academic Journal on Business Administration, Innovation, and SustainabilityLITY*, vol. 4, no. 04, pp. 53–69, Oct. 2024, doi: 10.69593/ajbais.v4i04.127.
- [16] M. Raman, "Mastering Change: The Journey from Waterfall to Agile and Hybrid Project Management," *Int J Res Appl Sci Eng Technol*, vol. 12, no. 11, pp. 1323–1332, Nov. 2024, doi: 10.22214/ijraset.2024.65371.
- [17] S. Gawande and S. Jiwani, "Constructive Effect of Hybrid Project Management Methodologies on Success Rates of Projects," *International Journal of Science and Research (IJSR)*, vol. 13, no. 10, pp. 50–54, Oct. 2024, doi: 10.21275/sr24929184750.
- [18] N. Bodke, K. Fatima Baig, A. Maurya, and A. Hatekar, "Navigating Project Management: A Comparative Analysis of Agile, Waterfall, and Hybrid Methodologies," *International Research Journal of Engineering and Technology*, 2025, [Online]. Available: www.irjet.net
- [19] I. Fatima, R. Nai, and R. Meo, "Machine Learning in Procurement with a View to Equity," in *Artificial Intelligence*, E. P. Dadios, Ed., Rijeka: IntechOpen, 2025, ch. 10. doi: 10.5772/intechopen.1008730.
- [20] H. S. da Cunha, F. L. C. Machado, X. L. Travassos Jr., and C. V. Ferreira, "Leveraging Artificial Intelligence in Contracting: A Digital Transformation for Public Institutions," *Journal of Bioengineering, Technologies, and Health*, vol. 7, no. 2, pp. 221–225, Jul. 2024, doi: 10.34178/jbth.v7i2.401.
- [21] A. Mavidis, D. Folinas, D. Skiadas, and A. Xanthopoulos, "Emerging Technologies Revolutionising Public Procurement: Insights from Comprehensive Bibliometric Analysis," *Adm Sci*, vol. 14, no. 2, Feb. 2024, doi: 10.3390/admsci14020023.
- [22] D. Edijala, S. Rakshit, and N. R. Vajjhala, "Applications of Artificial Intelligence in Public Procurement—Case Study of Nigeria," in *Computational Intelligence for Modern Business Systems: Emerging Applications and Strategies*, P. P. N. Chatterjee, S. Deepmala, K. Sandeep, and Draganand, Eds., Singapore: Springer Nature Singapore, 2024, pp. 513–522. doi: 10.1007/978-981-99-5354-7_27.
- [23] I. A. Olaleye, C. Mokogwu, A. Q. Olufemi-Phillips, and T. T. Adewale, "Optimizing procurement efficiency: Frameworks for data-driven cost reduction and strategic vendor management," *Magna Scientia Advanced Research and Reviews*, vol. 12, no. 2, pp. 164–171, Nov. 2024, doi: 10.30574/msarr.2024.12.2.0192.
- [24] M. Tan and W.-L. Lee, "Evaluation and Improvement of Procurement Process with Data Analytics," 2015. [Online]. Available: www.ijacsa.thesai.org
- [25] M. R. Sanda, M. I. Siminică, C. D. Avram, and L. Popescu, "Ghosts in the Machine: How Big Data Analytics Can Be Used to Strengthen Online Public Procurement Accountability," *Sustainability (Switzerland)*, vol. 16, no. 9, May 2024, doi: 10.3390/su16093698.
- [26] J. Yahaya, N. H. Z. Abai, A. Deraman, and Y. Y. Jusoh, "The implementation of business intelligence and analytics integration for organizational performance management: A case study in public sector," *International Journal of Advanced Computer Science and Applications*, vol. 10, no. 11, pp. 292–299, 2019, doi: 10.14569/IJACSA.2019.0101140.
- [27] N. Sakutemba, R. Muvungani, and T. Moyo, "Barriers to the Adoption of Electronic Procurement in Public Sector Construction Projects in Zimbabwe," *Journal of Construction in Developing Countries*, vol. 29, no. 2, pp. 231–255, 2024, doi: 10.21315/JCDC-12-23-0170.
- [28] I. Mohungoo, I. Brown, and S. Kabanda, "A Systematic Review of Implementation Challenges in Public E-Procurement," in *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, Springer, 2020, pp. 46–58. doi: 10.1007/978-3-030-45002-1_5.
- [29] T. McLaren and C. Middleton, "An Assessment of eProcurement Adoption Experiences in Canadian Purchasing Organizations," 2010. [Online]. Available: <http://aisel.aisnet.org/confirm2010/7>
- [30] N. Yahya and S. S. Maidin, "The Waterfall Model with Agile Scrum as the Hybrid Agile Model for the Software Engineering Team," in *2022 10th International Conference on Cyber and IT Service Management (CITSIM)*, IEEE, Sep. 2022, pp. 1–5. doi: 10.1109/CITSIM56380.2022.9936036.